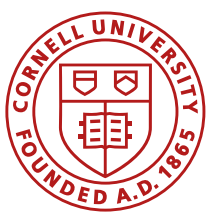
Introducing students to generative AI coding tools for reproducible projects

What I have learned so far

Ariel Ortiz-Bobea

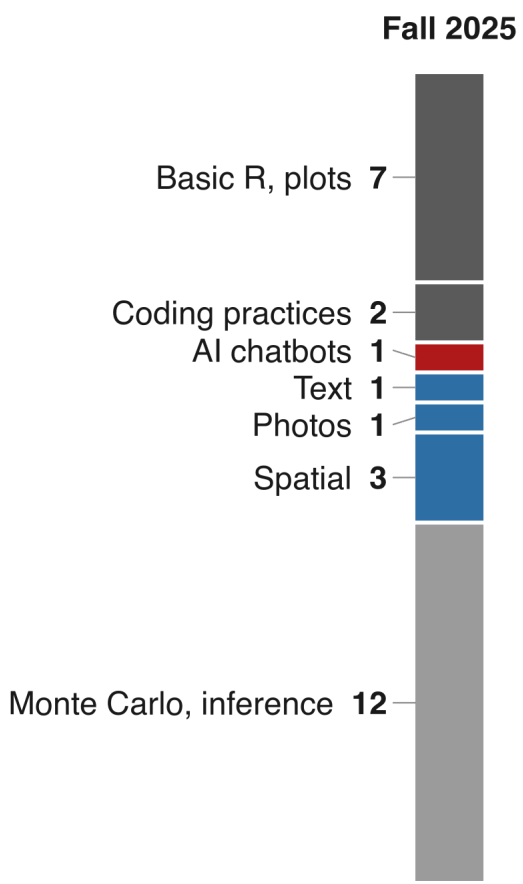
AERE Fall Virtual Workshop on Teaching and Pedagogy
AI Day, interactive demonstrations
November 10, 2026



Slides prepared with Claude Opus 5.5

The course through Fall 2025

A hands-on R course for MS students (AEM 6850), taught since 2019. Box heights count class meetings.

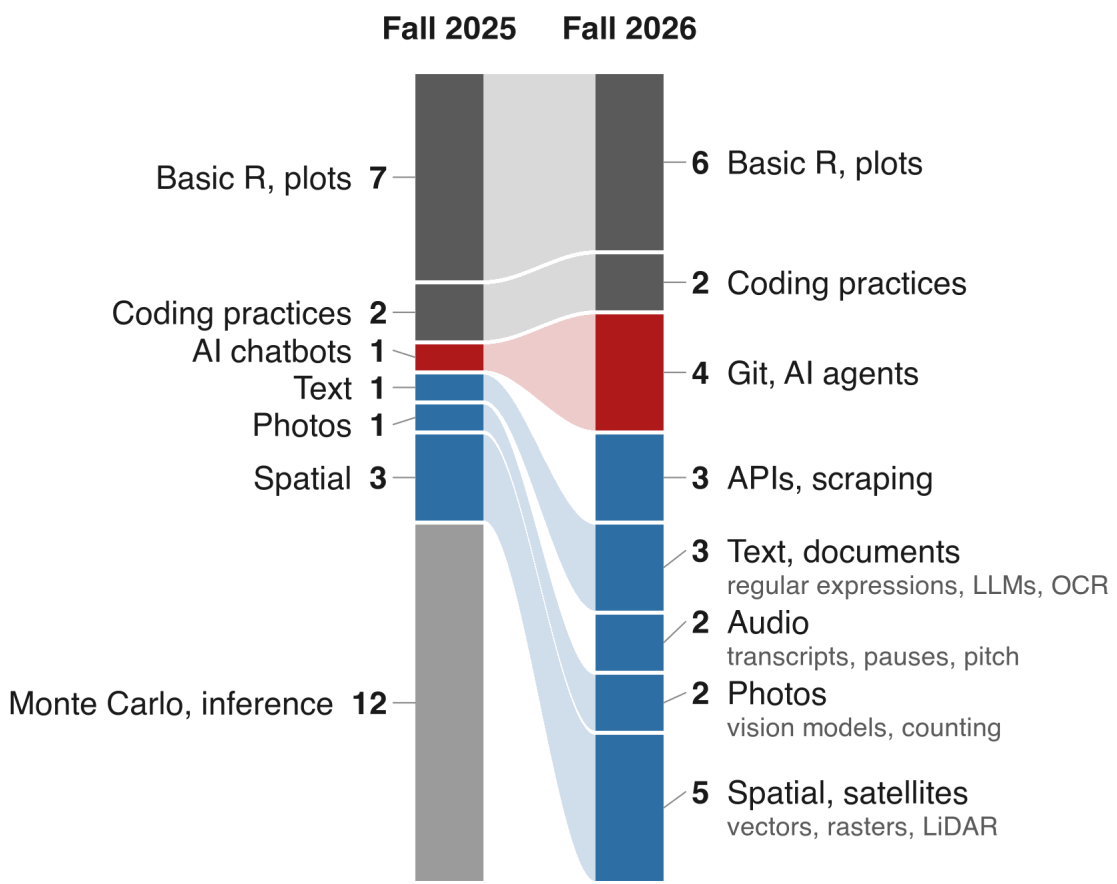


Learning goals

- Fluent in R
- Reproducibility, good coding manners
- A little version control
- Spatial data and a few other kinds
- Monte Carlo simulations to understand estimators
- TA graded homework by hand against my solution

The course in Fall 2026

Students would use AI anyway, so the course teaches it with its precautions.

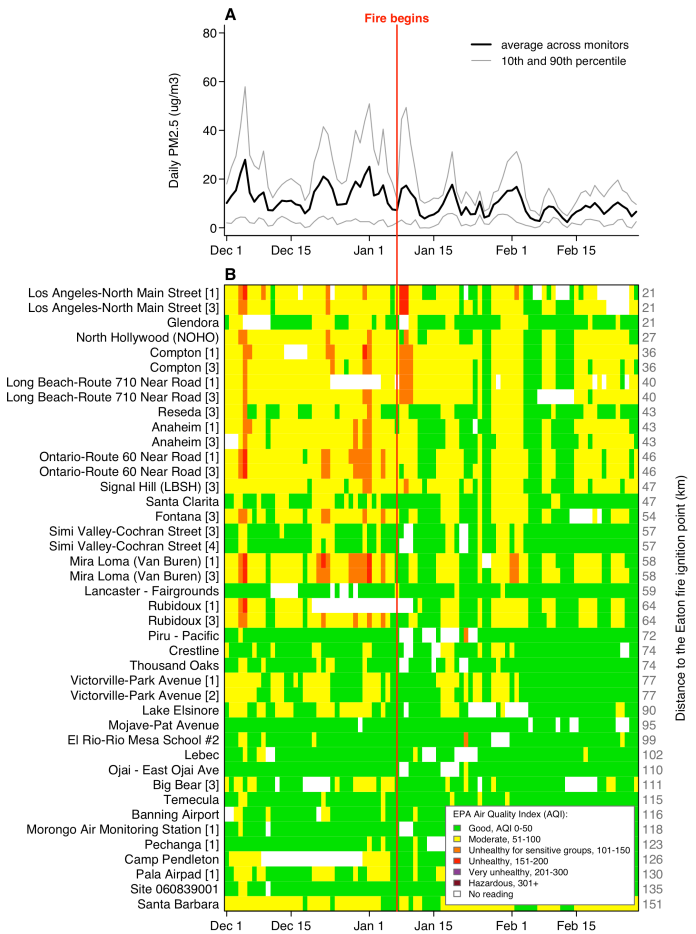


Learning goals

- Basic fluency in R
- **Coding with AI agents**, under version control
- **Text, audio and photos as data**, and maps
- **Far harder homework**, each one a simple application
- **Two AI coding agents grade**, and a third settles differences
- Human TAs review those grades instead of grading from scratch, and students can push back
- With fewer TAs, this is how the class scales past what I could grade alone

HW1: LA air quality in January 2025, the Eaton Fire

How bad was the air across the Los Angeles basin around the Eaton and Palisades fires?



- EPA's daily file, every monitor within 150 km of the Eaton Fire
- Each monitor's distance from the ignition point
- Three months, 3,995 rows, raw as EPA ships it
- No AI on this one
- HW2 then refactors this code with an AI agent and adds a wind panel

HW3: USDA's corn forecasts

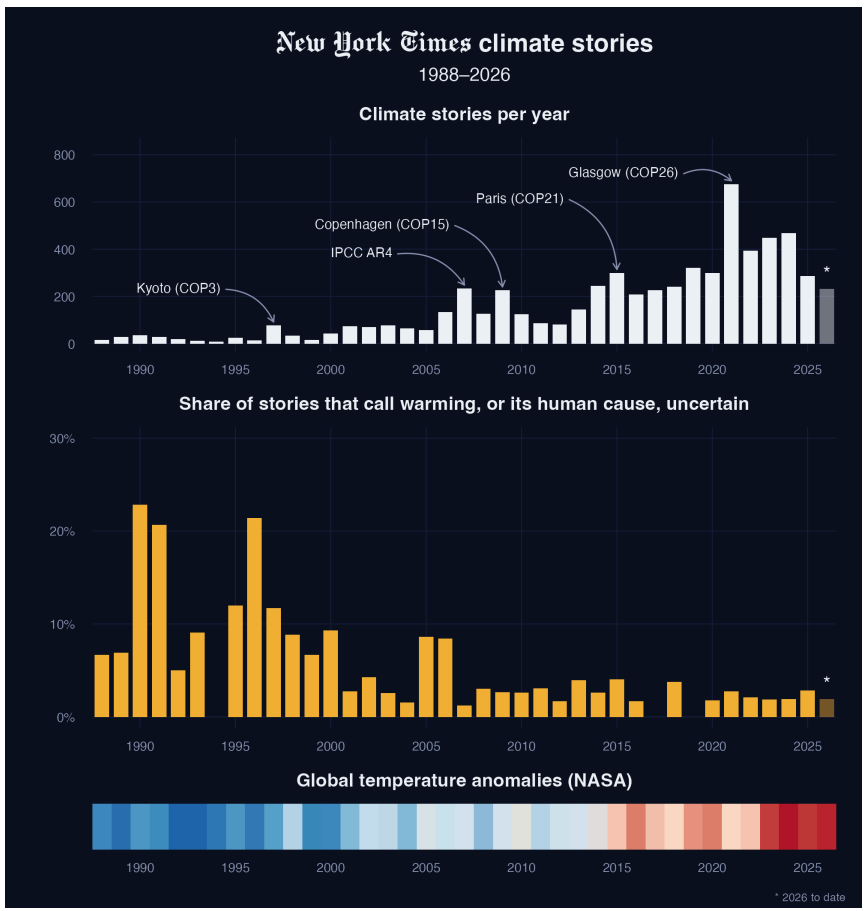
How did USDA's May forecasts of corn yield and price hold up, 2000 to 2026?



- **Goal**: use an API and regular expressions to approximate a target plot
- USDA's report archive, listed by an API
- 237 monthly reports read with regular expressions
- The world price of corn from FRED

HW4: climate stories that hedge on climate change and its human cause

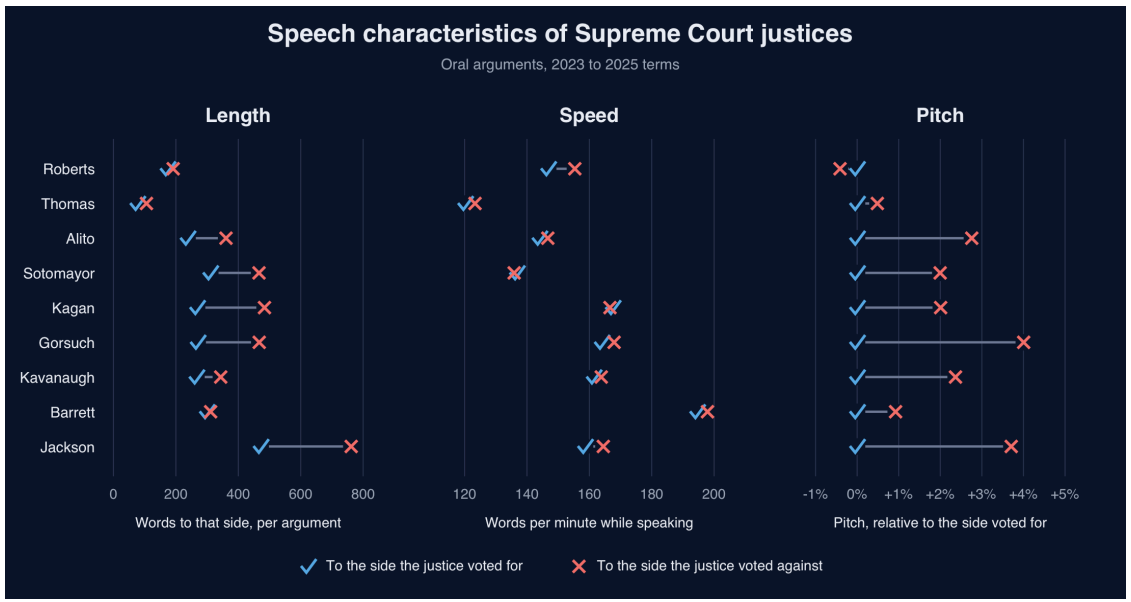
How often did New York Times climate stories call warming, or its human cause, uncertain?



- **Goal**: expose students to LLMs, and to how the labels vary with the prompt
- The Times' Archive API, 466 months since 1988
- An open-weights model on the laptop labels each story: Qwen 3.5, 9 billion parameters, run through Ollama
- A hand-coded study redone with a model, Boykoff and Boykoff (Global Environmental Change, 2004)
- About five hours on my laptop, downloads and model calls

HW5: the justices' voices

Do Supreme Court justices speak differently to the side they vote against?



- Oyez API: 183 arguments, 3.8 GB of audio, timed transcripts
- Votes from the Supreme Court Database
- Questions, speaking speed and pitch for each justice
- Builds on Epstein, Landes and Posner (Journal of Legal Studies, 2010) and Dietrich, Enos and Sen (Political Analysis, 2019)

How the rubric changed

Fall 2025, 9 points

Similarity 3
the figure looks like the target

Reproducibility 3
runs with no edits, a clear readme

Coding 3
sections, a parameter block, comments

Fall 2026, 15 points

Similarity 3
the figure looks like the target

Reproducibility 3
runs in a fresh clone and fetches its own data

Coding 3
sections, a parameter block, comments

Verification 3
checks that stop the script when the data is wrong

The note and the history 3
commits in order, choices explained, AI use disclosed

Scrape the PhD program placement page

In class, from an empty folder. The agent looks at the page, writes a script, runs it and reports.

1 Prompt

> Write code/placements.R. It reads the table "Recent job placements" at https://business.cornell.edu/admissions/graduate/phd-aem/for-hire/ and saves it as data/placements.csv with the page's four columns, year, name, position, organization, one row per graduate. Use the rvest package for the HTML and base R for the rest. Run it with Rscript code/placements.R; if it fails, fix the script and run again. Then show me the first five rows and the number of rows. Do not use git.

2 Code it wrote

code/placements.R, a new file of 38 lines (excerpt)

```
+# Scrape "Recent job placements" (Ph.D. in Applied Economics & Management)
+# from the Dyson "for hire" page and save it as data/placements.csv.
+library(rvest)
+page   <- read_html(url)
+tables <- html_elements(page, "table")
⋮
+write.csv(placements, out, row.names = FALSE)
```

3 Result

year	name	position	organization
2026	Trevor Dean Arnold	Postdoc	University of Calgary Department of Economics
2026	Molly Jane Doruska	Assistant Professor	Clarkson University
2026	Sijia Fan	Assistant Professor of Finance	University of Ottawa, Telfer School of Management
2026	Adrian Haws	Postdoc	Norwegian School of Economics
2026	Jihye Jang	Tenure-track Assistant Professor of Finance	Texas Christian University (TCU)

data/placements.csv, the first 5 of 109 rows

Add a postdoc column

A one-line change, and a count anyone can check with the browser's find box.

1 Prompt

> Add a column postdoc to the table before it is saved: TRUE when the position is a postdoc, FALSE otherwise. Change nothing else. Run it again and tell me how many rows are TRUE.

2 Code it wrote

code/placements.R, the diff

```
 rownames(placements) <- NULL

+# Flag postdoc positions; spellings on the page include "Postdoc", "Post-doc",
+# "Post Doctoral Fellow" and "Postdoctoral Associate".
+placements$postdoc <- grepl("post[- ]?doc", placements$position, ignore.case = TRUE)
+
 dir.create(dirname(out), showWarnings = FALSE, recursive = TRUE)
```

3 Result

year	name	position	organization	postdoc
2026	Trevor Dean Arnold	Postdoc	University of Calgary Department of Economics	TRUE
2026	Molly Jane Doruska	Assistant Professor	Clarkson University	FALSE
2026	Sijia Fan	Assistant Professor of Finance	University of Ottawa, Telfer School of Management	FALSE

25 of 109 rows are TRUE. On the page, find counts "postdoc" 15, "post-doc" 9 and "post doc" 1.

Add a country column

The same kind of prompt, another clean run.

1 Prompt

> Add a column country with the country of the organization. Change nothing else. Run it again and show me the number of rows per country.

2 Code it wrote

3 Result

year	name	position	organization	postdoc	country
2026	Trevor Dean Arnold	Postdoc	University of Calgary Department of Economics	TRUE	**Canada**
2026	Molly Jane Doruska	Assistant Professor	Clarkson University	FALSE	**United States**
2026	Sijia Fan	Assistant Professor of Finance	University of Ottawa, Telfer School of Management	FALSE	**Canada**

Its reply: “The 32 non-US assignments are printed above and all look right.”

Add a country column

The same kind of prompt, another clean run.

1 Prompt

> Add a column country with the country of the organization. Change nothing else. Run it again and show me the number of rows per country.

2 Code it wrote

code/placements.R, the diff: 32 new lines (excerpt)

```
+# Country of the organization. The page does not list it, so it is inferred
+# from the organization name: the first matching pattern below wins, and
+# anything unmatched is treated as United States (the large majority).
+country_patterns <- c(
+  "Australia"    = "Monash|New South Wales|University of Sydney",
+  "Canada"       = "Alberta|Calgary|Ottawa",
+  "Germany"      = "Ludwig-Maximilian|DTUM",
⋮
+)
+placements$country <- "United States"
```

3 Result

year	name	position	organization	postdoc	country
2026	Trevor Dean Arnold	Postdoc	University of Calgary Department of Economics	TRUE	**Canada**
2026	Molly Jane Doruska	Assistant Professor	Clarkson University	FALSE	**United States**
2026	Sijia Fan	Assistant Professor of Finance	University of Ottawa, Telfer School of Management	FALSE	**Canada**

Its reply: “The 32 non-US assignments are printed above and all look right.”

AI agents hard-coded the answer; the proof the lesson landed is the audible gasp.

How I envision the course will change

1. **In-class quizzes**
   To verify basic knowledge and judgement.

2. **More emphasis on creativity and economic thinking**
   AI is a new means to answer these questions with novel types of data and models. It augments students' thinking and empowers them to ask their own questions.

3. **Student-led projects**
   Provide hands-on experience for students to pursue their own ideas.

4. **Enhance student feedback**
   Automate how I hear from students beyond course evaluations, on what they like and where they struggle. I still need to think this through.